

## **ULTRA-HIGH-PERFORMANCE FIBER-REINFORCED CEMENTITIOUS COMPOSITE OCTET LATTICE UNIT CELLS**

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### **1. INTRODUCTION**

#### **1.1. Motivation and Impact**

Concrete is the most prolific building material in the world today due to its versatility and widespread availability, yet it is far from an ideal material; it is heavy, weak in tension, and a poor insulator. Beyond the material properties, concrete has a significant environmental impact due to its high carbon footprint and demand for raw materials. As human population increases and more people trend toward living in urban areas [1], the need for fully optimized materials and structures becomes increasingly important to improve the performance of buildings and reduce their environmental impact.

#### **1.2. Cellular Solids**

Lattices are used in nature to great effect. With structures tailored to suit the material, excellent mechanical properties are achieved using a minimal amount of material. Advancements in additive manufacturing in recent years have enabled the creation of three dimensional lattices optimized for the base material. This yields excellent mechanical and thermal properties, and simultaneously reduces the amount of material used [2]. An ideal lattice for concrete is one that places the material primarily under compression, thus minimizing the influence of concrete's poor tensile strength. One such cell structure that is shown to have this characteristic is the octet lattice. Previous studies have shown that the octet lattice approaches linear relation between strength and density for metal and polymer base materials [3], yet its effectiveness for quasi-brittle materials such as concrete or cement composites remains unknown.

### **2. EXPERIMENTAL PROGRAM**

#### **2.1 Focus of the Study**

Cellular concrete is traditionally achieved by foaming cement with aluminum powder resulting in a random pore structure. Concrete and cement mortars foamed in this way demonstrate strength decreasing at a cubic proportion to density. This study investigates the relationship between compressive strength and effective density for octet lattice unit cells made from an ultra-high-performance fiber-reinforced cementitious composite (UHP-FRC) and compares the results to those of ultra-high-performance cement mortar (UHPC) foamed with conventional methods.

## 2.3 Methods

To form the complex octet lattice structure, molds were first manufactured out of acrylonitrile butadiene styrene (ABS) using fused deposition modeling (FDM). Molds were then infiltrated with the UHP-FRC, and after seven days the mold material was dissolved away in an acetone bath. Following removal of the molds, lattices were cured in a fog room for twenty-one additional days and then tested in compression.

A true lattice material consists of many unit cells, however for this study only single unit cells were tested. During testing, unit cells were placed in aluminum holders intended to mimic the boundary conditions imposed by a repeated lattice.

## 2.2 Results

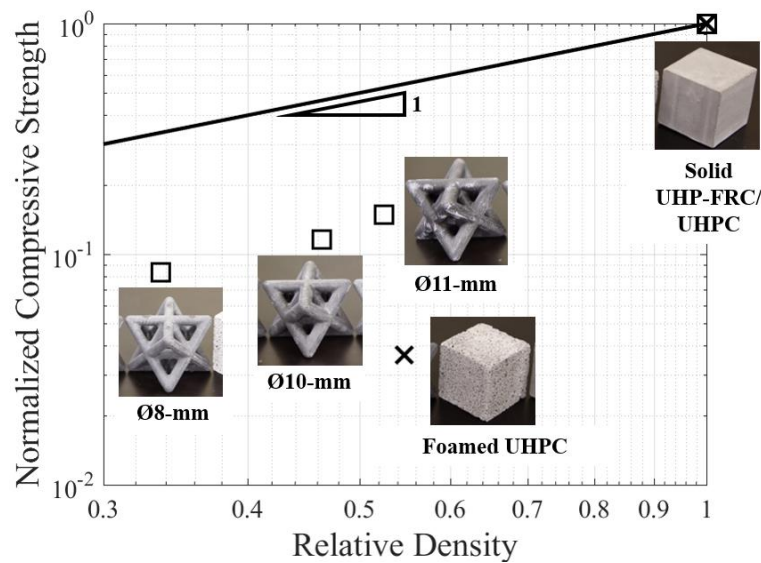


Figure 1: Normalized compressive strength vs. relative density on log-log scale

Three variations of the octet unit cell with member diameters of 8-mm, 10-mm, and 11-mm were tested in this study, and one conventionally foamed mortar for comparison. Figure 1 shows the normalized compressive strength vs. relative density of the UHP-FRC octet trusses and the foamed UHPC. Strengths for each specimen have been normalized to the strength and density of a solid cube of the material used (foamed mortar specimen is normalized by the strength of same mortar without foaming).

Results of this study show that octet lattices can achieve 40% lower density than foamed UHPC, and achieves an improved relationship between strength and relative density. While further optimization of the size and scaling of the lattices as well as improved methods of manufacturing are required to make it feasible, these results demonstrate the improved mechanical behavior at low densities when using an octet lattice compared to random porosity.

## REFERENCES

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